

MR BRAIN PERFUSION WITH AND WITHOUT CONTRAST

Results

Study Result

Narrative & Impression

UNIVERSITY HEALTH

4502 MEDICAL DRIVE, SAN ANTONIO, TX 78229

PH: 210-358-8532

PROCEDURE(S): MR BRAIN PERFUSION WITH AND WITHOUT CONTRAST

ACCESSION(S): UH5837637

DATE OF EXAM: 1/24/2026 7:51 PM

REFERRING PHYSICIAN: MAHMOUD EL HAJJ

EXAM: MRI brain with and without IV contrast.

TECHNIQUE: Multiplanar, multisequence MRI of the brain was performed both before and after the administration of intravenous gadolinium contrast.

CONTRAST: 12 mL Gadavist

HISTORY: differentiate tumor progression vs. radiation necrosis

Additional history on chart review: s/p SRS x5 to brain lesions 4-5/2023

COMPARISONS: CT head 6/5/2025. Outside facility MR brain 1/16/2026, 11/12/2025, 8/8/2025

FINDINGS:

Brain parenchyma and contrast enhancement:

* Slightly increased size of frontal lobe lesion measuring up to 1.5 cm, previously measuring up to 1.4 cm on 1/16/2026. Unchanged adjacent T2/FLAIR hyperintensity. This was measured on the T2 imaging secondary to lack of postcontrast imaging on the immediate prior study. Increased cerebral blood flow on the ASL sequence. Intermediate elevated cerebral blood volume within the enhancing portion of the lesion.

* Redemonstration of lesion in the midline cerebellar vermis measuring 1.4 x 2.4 x 1.3 cm, similar to slightly increased in size compared to 1/16/2026 given limitations of lack of IV contrast on that exam. Lesion again shows susceptibility artifact suggesting associated hemosiderin deposition. Unchanged adjacent T2/FLAIR hyperintensity. Lesion is outside the field of view on ASL sequence. Associated intermediate increased

cerebral blood volume in the enhancing portion of the lesion.

* New punctate 4 mm enhancing lesion in the left inferior cerebellum (1402:51).
Evaluation for perfusion could not be performed given small size of the lesion.

No acute infarct or acute intraparenchymal hemorrhage. There is superimposed mild chronic microangiopathic ischemic changes of the supratentorial white matter.

Ventricular system: No hydrocephalus or midline shift.

Extra axial spaces: No epidural, subdural, or subarachnoid hemorrhage. No basal cistern effacement.

Vascular system: No loss of the major vascular flow voids.

Visualized paranasal sinuses: No mucosal thickening. The bilateral mastoid air cell complexes are clear.

Visualized orbits: No ocular or retrobulbar pathology.

Calvarium and skull base: No osseous destruction.

Pituitary and Pineal Glands: No mass lesion.

Soft tissues: No significant abnormality.

IMPRESSION:

1. Left frontal lobe metastasis has slightly increased in size on T2-weighted imaging when compared to the 1/16/2026 exam. There is evidence of peripheral hypercellularity, increased relative cerebral blood flow, and indeterminate increase in relative cerebral blood volume. Findings favor progressive metastasis over radiation necrosis.
2. Inferior cerebellar treated metastasis with hemosiderin staining appears grossly similar to the prior 1/16/2026 exam and subtle increase in size from 11/12/2025 exam. There is intermediate increased relative cerebral blood volume. However, presence of blood products may limit evaluation. This could represent recurrent/residual neoplasm versus radiation necrosis.
3. New left inferior cerebellar hemisphere 4 mm enhancing lesion concerning for new metastasis and interval progression of metastatic disease.
4. Unchanged left frontal lobe and cerebellar vermis vasogenic edema. No midline shift or herniation syndrome.

I, Jason Lally, have personally interpreted the image(s) and dictated this examination or I have reviewed the image(s) as well as the residents interpretation and agree with

the interpretation.

Final Signed by Jason Lally, on 1/26/2026 2:43 PM

Images

Images

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